

FINAL EXPLANATION: MATHEMATICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: MATHEMATICS GRADE 10 – 3.3 VECTORS I: FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

Driving question: Why did the Kisumu ferry miss Mbita, and how can mathematics give the captain the tools to always reach the right destination, no matter the current or wind? Use what you learned in Lessons 1 to 8 to answer each part below. Write in full sentences, show every step of your working, and include units. Draw neat labelled diagrams on grid paper where asked. Remember the situation: a ferry leaves Kisumu port to travel due east to Mbita, 60 km away. The engine pushes it east at 20 km/h, while a Lake Victoria current pushes it south at 8 km/h.

Part 1: Scalars, vectors and why direction matters

(a) Name two quantities from the ferry story. Say which is a scalar and which is a vector, and explain how you decided. (b) Give one more example of a scalar and one more example of a vector from Kenyan life (for example Kericho, Mombasa or a matatu trip from Nairobi to Nakuru). (c) The captain said, 'My engine pointed at Mbita, so I should have arrived at Mbita.' Using the words scalar and vector, explain what the captain forgot.

Write your answer here:

Part 2: Writing and adding the vectors

(a) Write the engine thrust and the lake current as column vectors, using east as the positive horizontal direction and north as the positive vertical direction. (b) Write the position vector of Mbita relative to Kisumu port (the origin) for the captain's intended journey of 60 km east. (c) Add the engine and current column vectors to find the resultant vector. (d) Describe how you would show this addition using the triangle law, and say how the parallelogram law gives the same result. (e) Explain why the resultant vector has a negative bottom entry.

Write your answer here:

Part 3: How fast and in what direction did the ferry really go?

(a) Calculate the magnitude of the resultant vector. Show the formula you use and give your answer to 1 decimal place. (b) Calculate the angle between the resultant path and due east, and state it in words such as 'south of east'. (c) The ferry needs 3 hours to cover 60 km at its engine speed of 20 km/h. Calculate how far south the current pushes it in that time. (d) A classmate says: 'The ferry's speed over the lake is $20 + 8 = 28$ km/h.' Explain why this is wrong.

Write your answer here:

Part 4: Correcting the course

(a) Suppose a ferry has landed at a wrong point L and needs to reach the intended point M. Explain in words how you find the correction vector from L to M using position vectors. (b) Explain what the negative of a vector is, and what you get when you add a vector to its negative. (c) The captain wants to keep the engine speed at 20 km/h but tilt the heading north of east so that the northward part of the engine velocity exactly cancels the southward current. Find the angle north of east for the heading, and say what this tells the captain to do differently from before. Give your answer to 1 decimal place.

Write your answer here:

Part 5: What if the current gets stronger? Answering the driving question

(a) Suppose the current doubles. Write the new current as a column vector using scalar multiplication, and say whether it is parallel to the original current. How do you know? (b) Calculate the new resultant column vector and its magnitude when the engine is still (20, 0). (c) Use your results from Parts 1 to 4 to write a short paragraph (4 to 6 sentences) answering the driving question: why did the ferry miss Mbita, and how does vector mathematics help the captain always reach the right destination?

Write your answer here:

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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Scalar and vector reasoning	Correctly classifies quantities as scalars or vectors with clear reasons, and explains fully how treating velocity as a scalar caused the ferry to miss Mbita.	Classifies most quantities correctly and gives a reasonable explanation of why direction matters in the ferry story.	Confuses scalars and vectors or does not link direction to the ferry missing Mbita.
Vector representation and addition	Writes all column and position vectors with correct signs, adds them accurately, and explains both triangle and parallelogram laws.	Writes and adds the column vectors correctly with minor sign or explanation slips.	Makes frequent errors in column vector signs or addition, or cannot describe the addition laws.
Magnitude, direction and calculation	Uses Pythagoras and arctan accurately with units and sensible rounding (about 21.5 km/h, 21.8° south of east, 24 km drift, 23.6° correction), and explains why $20 + 8$ is wrong.	Calculates most values correctly with working shown, with small rounding or unit errors.	Cannot set up the right-angled triangle or calculate magnitude and angle correctly.

Correction and scalar multiplication	Explains subtraction of position vectors, the negative and zero vector, and parallel scalar multiples, and applies them correctly to the corrected heading and doubled current.	Applies most of these ideas correctly with some gaps in explanation.	Does not connect subtraction or scalar multiplication to correcting the ferry's course.
Communication and link to the driving question	Writes a clear, connected answer to the driving question using correct vocabulary, labelled diagrams and Kenyan context.	Answers the driving question with mostly correct vocabulary and some connection of ideas.	Answer is incomplete or lists facts without linking them to the ferry phenomenon.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

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